Applied Engineering & Robotics — Syllabus

School Year: 2026–2027
Instructor: Matt Bombich
Email: mbombich@vicksburgschools.org
Room: 70
Office Hours: Monday–Thursday, 2:45–3:15 pm
Prerequisites: None — basic math and science helpful.

Course Description

Applied Engineering & Robotics is a full-year, project-based course where you design, build, and program a real autonomous robot. You will use professional tools — Autodesk Fusion CAD software, a Raspberry Pi Pico 2W microcontroller, MicroPython, 3D printers, and CNC machines — to engineer L.E.O. (Logic, Electronics, and Operations), a custom robotics platform built from the ground up.

Work follows the Engineering Design Process: you do not just assemble kits. You model every component in CAD, iterate based on testing and measurements, write code that controls motors and sensors, and document your decisions in a digital portfolio. By the end of the year you will have a portfolio of engineering work you can use in future courses, job applications, and college admissions.

Academic Credits: 4th year related math / 3rd year science
College Credit: Pending KVCC articulation agreement

Course Objectives

By the end of the year, you will be able to:

- Apply the Engineering Design Process to define problems, generate solutions, and iterate based on evidence.
- Model mechanical components in Autodesk Fusion 360, produce technical drawings, and manage a multi-part assembly.
- Program a Raspberry Pi Pico 2W in MicroPython to control motors, read sensors, and implement autonomous behaviors.
- Fabricate parts using 3D printers, a CNC router, and hand tools — with precision verified by calipers.
- Wire and troubleshoot electronic circuits including motor drivers, IR sensors, and ultrasonic distance sensors.
- Practice shop safety: PPE, 5S, lockout/tagout, and safety data sheets.
- Reflect on your engineering decisions in a digital portfolio and connect your work to career pathways.

Materials & Tools

All materials and tools are provided by the school. You do not need to purchase anything.

Category	Items
Electronics	Raspberry Pi Pico 2W, L.E.O. interface board, IR sensors, HC-SR04 ultrasonic sensors, servo motor
Software	Autodesk Fusion (CAD), Thonny IDE (MicroPython), Google Workspace
Fabrication	Bambu and Creality 3D printers, CNC router, drill press, hand tools, calipers

Safety	Safety glasses, gloves, hearing protection (provided)
Resources	Class portfolio site, Google Classroom, Pico 2W reference docs

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Grading

Category	Weight
Portfolio Deliverables	50%
Quizzes	20%
Midterm & Final Exam	20%
Safety, 5S & Participation	10%

Portfolio Deliverables (50%) are the core of your grade. Every major project milestone — your Design Brief, each CAD component, safety certifications, programming assignments, and reflections — is submitted through your digital portfolio at mbombich-robotics.github.io/CTE. Each deliverable has a rubric visible in the portfolio before you submit. Point values reflect the complexity of the work.

Quizzes (20%) are given at the end of major units to assess understanding of content and skills covered.

Midterm & Final Exam (20%) are cumulative assessments given at the end of each semester.

Safety, 5S & Participation (10%) — You begin each semester with 50 points. Ten points are deducted per incident, including: not wearing required PPE, leaving tools or materials out after use, failing to participate in 5S cleanup, or other unsafe behaviors. Serious or repeated violations are referred to administration regardless of remaining points.

Unit & Pacing Overview

Unit	Title	Approx. Weeks	Key Skills
1	Engineering Design Process	1–2	EDP steps, design brief, problem definition, decision matrix
2	CAD & 3D Printing	3–9	Fusion 360 modeling, technical drawings, caliper measurement, iterative printing (C1–C7)
3	Shop Safety	10	Drill press, CNC, PPE, LOTO, SDS — certification required before build phase
4	Programming & Robotics	11–25	MicroPython, motor control, sensors, autonomous navigation with L.E.O.
5	AI & Machine Learning	26–28	Intro ML concepts, computer vision basics, ethical implications
6	Career Readiness	ongoing	Career research, paycheck analysis, portfolio reflection

Exact pacing is posted in Google Classroom. Weeks are approximate and may shift based on build and testing time.

Second-Year Students

Students returning for a second year work at Level 2 (B-track) or Level 3 (A-track). You will complete extension projects, advanced programming challenges, and additional independent work while your classmates cover foundational content. Specific expectations are discussed individually at the start of the year.

Student Leadership

FIRST Robotics is available to students at all levels as an opportunity to apply engineering skills in a competitive team environment and develop teamwork, time management, and professional communication. The Capstone Project includes a formal presentation of your engineering work to a real audience.

Classroom Policies

Safety — Non-Negotiable
PPE is required whenever shop tools or electronics are in use. A safety violation results in a 10-point deduction from your Safety, 5S & Participation grade and immediate removal from the work area. Repeated violations will be referred to administration.

Attendance
This class is hands-on — you cannot make up lab time from home. Excused absences receive full make-up opportunities.

Late Work
Portfolio deliverables are accepted up to 3 school days late with a 10% per day penalty. After 3 days, work is accepted for partial credit at the teacher's discretion. Talk to Mr. Bombich before the deadline if you need an extension.

Academic Integrity
You may use AI tools to help you understand concepts, debug code, or generate ideas — but the work you submit must reflect your own thinking. Copying another student's code, design, or written reflection is academic dishonesty and will be handled per school policy.

Accommodations
Accommodation plans are fully supported. Please discuss your needs privately and early — accommodations work best when we plan ahead together.

Contact & Resources

Portfolio	mbombich-robotics.github.io/CTE
Class Website	mbombich-robotics.github.io/CTE/hub.html — all assignments, rubrics, pacing, and announcements
Email	mbombich@vicksburgschools.org — replies within 24 hours on school days
Tutorial	Tuesdays & Thursdays — request your assignment by end of day Monday or Wednesday. Students with missing work are assigned automatically.